MINING DEALER BEST PRACTICE

Cooling System Performance for LWL 994

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | MARC Management

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1.0 Introduction

The purpose of this Best Practice is to establish procedures within preventive maintenance to avoid unexpected downtimes or damage in major components due to high engine, transmission or implement temperatures. When the radiator is externally obstructed (dirt from the front loader plus water or oil) the cooling system loses efficiency and also if there is internal contamination in the cooling system.

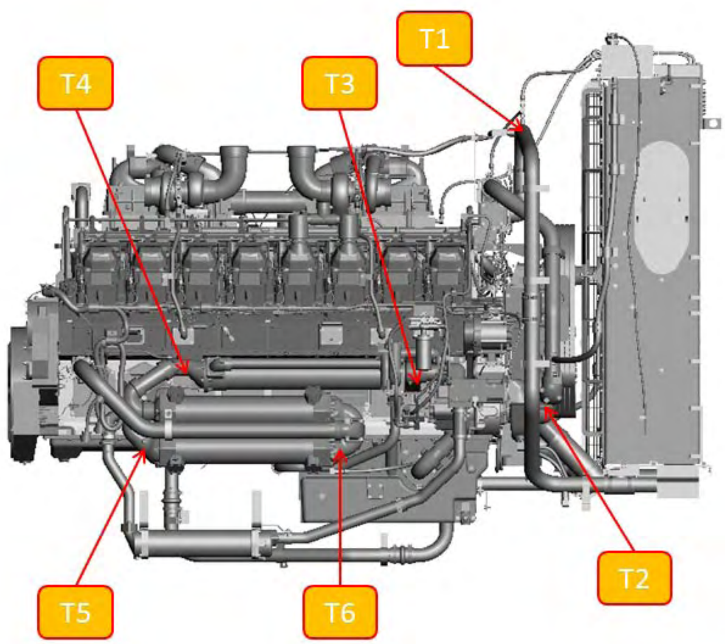
With periodic evaluations and inspections of the cooling system we will:

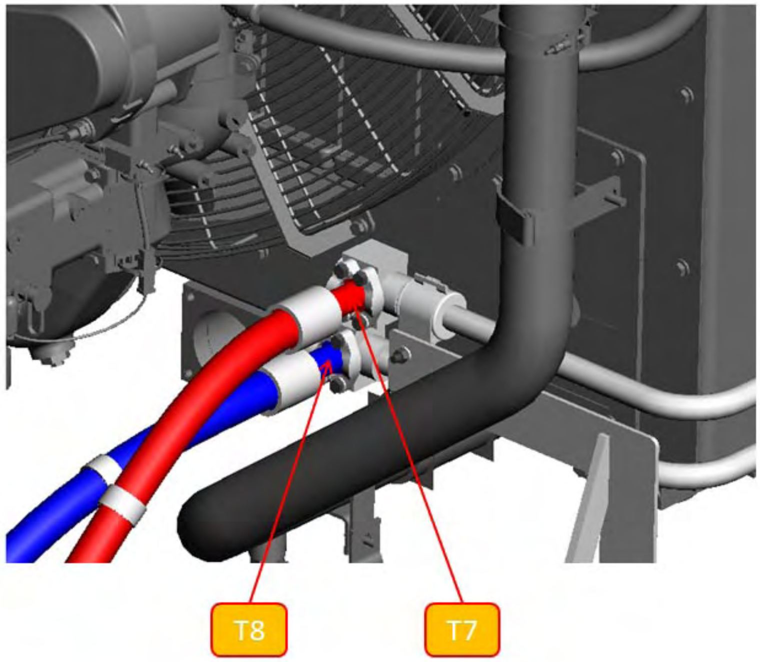
- Rule out problems related to dirt and clogging of radiator and coolants
- Avoid catastrophic failure associated to high temperature caused by contamination of coolants and radiators
- Determine whether the coolants and radiator are dissipating the heat correctly

2.0 Description of Best Practice

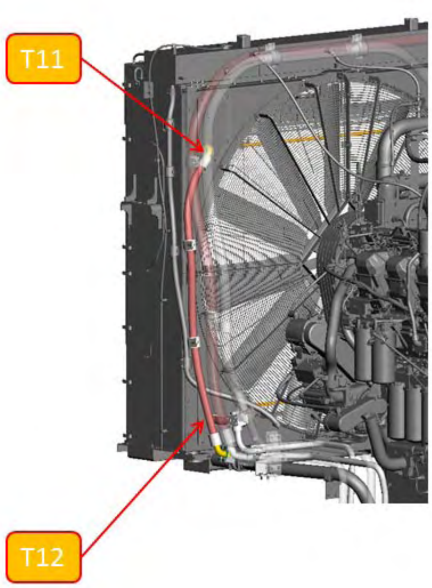
Temperature measuring procedure:

The following graphics show engine coolant diagrams and the locations to be evaluated by temperature.

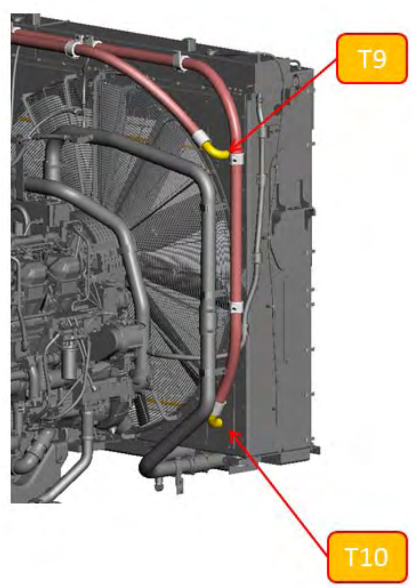




Machine - Left



Machine - Right



Machine - Left

Temperatures Chart

System	Temp Location (On Diagram)	Referenced Value	Measured Value	Expected Tolerance
Radiator Inlet	T1	107.9		
Radiator Outlet	T2	99.5		
Radiator Delta	**T2-T1**	**-8.4**	**0**	+/- 1.0°C
Engine Oil Water Cooler Inlet	T3	99.5		
Engine Oil Water Cooler Outlet	T4	101.4		
Engine Oil Cooler Delta	**T4-T3**	**1.9**	**0**	+/- 1.0°C
Powertrain Water Cooler Inlet	T5	100.5		
Powertrain Water Cooler Outlet	T6	103.3		
Powertrain Cooler Delta	**T6-T5**	**2.8**	**0**	+/- 1.0°C
Brake Cooler exempt due to high operator variability and minimal system impact				
Powertrain Oil-to-Air Inlet	T7	127.4		
Powertrain Oil-to-Air Exit	T8	101.1		
Powertrain Oil-to-Air Delta	**T8-T7**	**-26.3**	**0**	+/- 4.0°C
Implement Cooler Inlet	T9	94.7		
Implement cooler Exit	T10	79		
Implement Sys Delta	**T10-T9**	**-15.7**	**0**	+/- 2.0°C
Steering Cooler Inlet	T11	80.4		
Steering Cooler Exit	T12	68.1		
Steering Sys Delta	**T12-T11**	**-12.3**	**0**	+/- 2.0°C
All temperature deltas' referencing traditional machine truck loading cycles. Deviation from truck loading cycles will results in inaccurate data. Fully opend thermostate is also required which equates to a coolant temperature at or above 92°C				

Air flow test procedure:

This procedure is to be used as **REFERENCE ONLY** relative to supporting historical data from the machine being tested. Machine cooling system should be set to factory specs and pressures and cooling cores completely cleaned.
Use this data as a reference for future test for the machine S/N being tested.

The procedure of the air flow test with 8T-2700 tool and pressure registry of the coolant system is described below.

It is for measuring the amount of air passing through the radiator.

- Put the machine in a horizontal terrain
- Start the engine and park the equipment
- Locate a ladder in the posterior part of the equipment and secure it
- Configure the blow by 8T-2700 equipment for measuring
- Measure 8 points (shown below)
- To measure air pressure in the same point, mark the radiator's guards with a marker
- Engine rpm and fan rpm should be registered to obtain a standard measurement

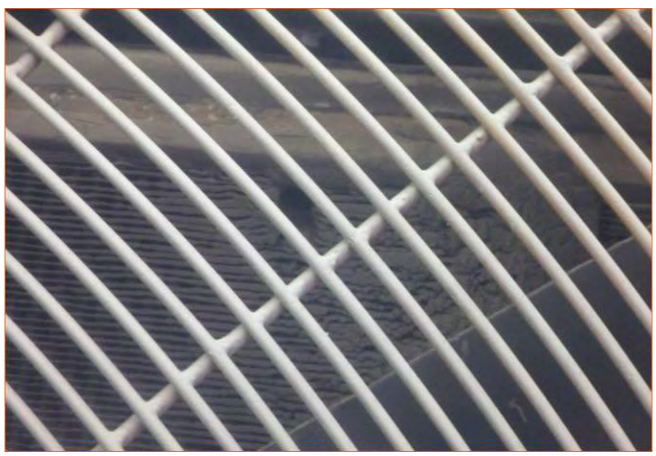
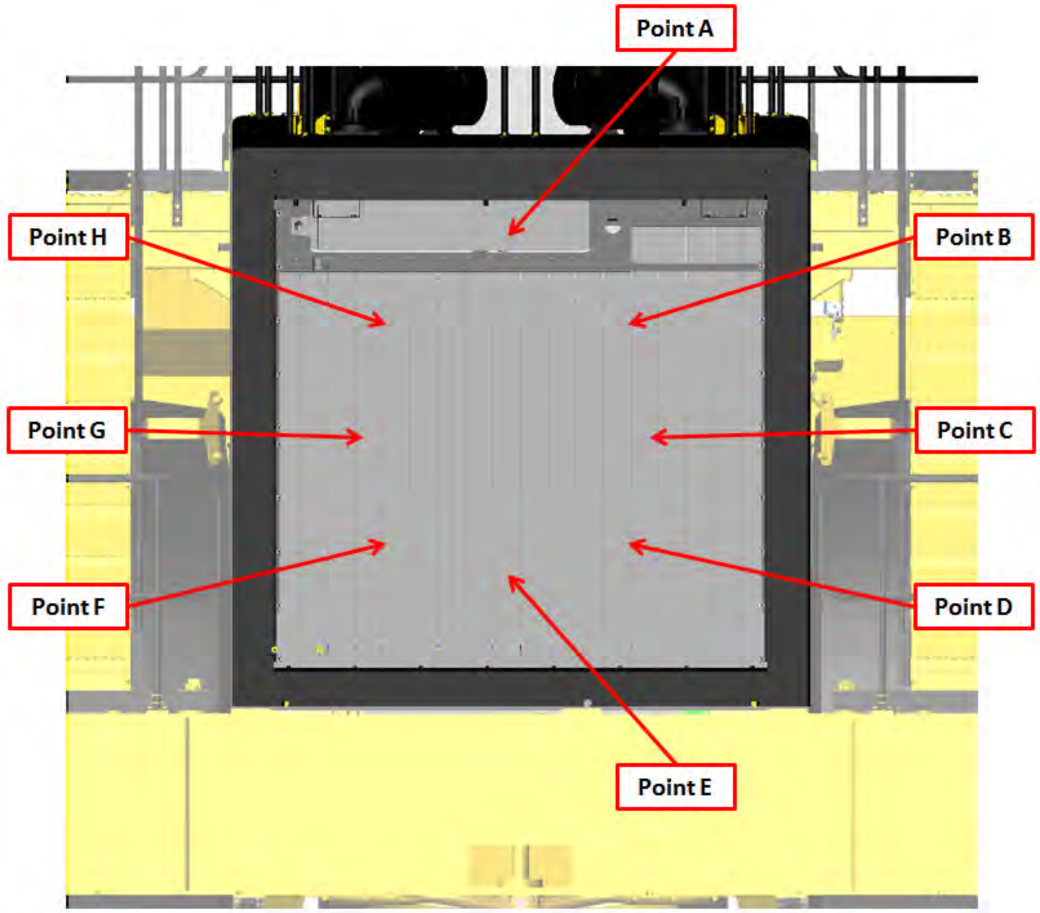


Image 1- Exterior clogging of the implement's core

The information must be saved in order to generate particular trends of each machine.
Repair actions, such as cleanliness, parts replacement, etc. can be generated depending on the analysis.

The data analysis compared to previous experiences may indicate that less air amount is going through the radiator, impacting directly on the cooling system performance.

Reference Example:



Machine - Rear (Exit Grills Removed for Clarity)

S/N: 442-XXXXX				
	Airflow Reading #1 (Machine Baseline)	**Airflow Reading #2**	**Airflow Reading #3**	**Airflow Reading #4**
Point A				
Point B				
Point C				
Point D				
Point E				
Point F				
Point G				
Point H				
NOTE: Reading #1 should be taken after cooling system is thouroghly cleaned. Reading #1 should then be used as a baseline point for all future measurements pertaining to the same machine serial number				

3.0 Implementation Steps

a. Report about the process to those in charge of the machine
b. Ensure the machine is safe to approach
c. Complete Pre Checks
    i. Check for proper coolant level
    ii. Check/replaced thermostats – ensure proper orientation
    iii. Check water pump for functionality
    iv. Check for debris build-up in the airstream of ALL cooling cores – clean as needed
d. Establish who will be responsible of taking data
e. Periodically take data
f. Complete Post Checks
    i. Replace any part removed
    ii. Double check for leaks

4.0 Benefits

The high engine temperature can cause a catastrophic engine failure if not controlled. The cost of a 3516B engine of a 994F is $570,253. Downtime of an engine failure is high because of the labor involved and the probability of finding an available engine when needed.
In Glencore Tintaya mining site, at 4100 mosl, we've had two catastrophic engine failures which represent long downtime periods due to coolant's high temperature. Also, when we have very high coolant temperatures for a long period of time, there is also the possibility of compromising other systems such as the converter, which also uses coolant.

5.0 Resources Required

- Staff involved in the job
- Ladder
- Inspector or supervisor

6.0 Supporting Attachments / References

- SEBD0518 "Know Your Cooling System{0708, 1000, 1350, 7000}"

7.0 Acknowledgements

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